N17 Centre action

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1

The classical equations of motions are

$$\dot{q} = \frac{\partial H}{\partial p}, \qquad \dot{p} = -\frac{\partial H}{\partial q} \tag{1}$$

we wish to find the action generating function/centre action $S^{(3)}(\bar{p}, \bar{q})$, which satisfy

$$\Delta q = \frac{\partial S^{(3)}}{\partial \bar{p}}, \qquad \Delta p = -\frac{\partial S^{(3)}}{\partial \bar{q}} \tag{2}$$

where $\bar{q}$ and midpoint position $(q + q')/2$ and $\Delta q = q' - q$ is the displacement due to some time evolution t, similarly $\bar{p} = (p + p')/2$ and $\Delta p = p' - p$. Appendix A shows the following for a harmonic oscillator

$$S^{(3)}(\bar{p}.\bar{q}) = \frac{2}{\omega} \tan\left(\frac{\omega t}{2}\right) H(\bar{p}, \bar{q}) \tag{3}$$

notice the resemblance of the pre factor with end to end fluctuation of open imaginary time path. (we will not focus on this for now) For small $t = \Delta t$, $\tan\theta \simeq \theta$ and we have $S^{(3)} \simeq H(\bar{p}, \bar{q})\Delta t$ and substitute back in Eq.(2) gives

$$\frac{\Delta q}{\Delta t} \simeq \frac{\partial H(\bar{p}, \bar{q})}{\partial \bar{p}}, \qquad \frac{\Delta p}{\Delta t} \simeq -\frac{\partial H(\bar{p}, \bar{q})}{\partial q} \tag{4}$$

and we recover Eq.(1) as $\Delta t \to 0$.

Now consider extended phase space inhibited by $\mathbf{r} = (q, p, t)$, with phase velocity $\mathbf{u} = (\dot{q}, \dot{p}, 1)$. It we forget all we know about classical mechanics including Eq.(1), and postulate that

$$\nabla \cdot \mathbf{u} = 0 \tag{5}$$

much like a magnetic field is divergence free, we can find a vector potential, we also find a vector potential $\boldsymbol{\theta}$ where

$$\mathbf{u} = -\nabla \times \boldsymbol{\theta} \tag{6}$$

where $\nabla = (\partial_q, \partial_p, \partial_t)$ is the extended phase space gradient. Since $\nabla \times \nabla f = 0$ for scalar function f, $\boldsymbol{\theta}' = \boldsymbol{\theta} + \nabla f$ would give the same $\mathbf{u}$, this is gauge invariance. Upon choosing $f = -\int dp\, \theta_p$, we have $\boldsymbol{\theta} = (p, 0, -H)$, then it is easy to show this recovers Hamilton's equations. Define action as the following path integral

$$S_\gamma = \int_\gamma \boldsymbol{\theta} \cdot d\mathbf{r} \tag{7}$$

where γ is a path in extended phase space. However, since the choice of $\boldsymbol{\theta}$ is not unique, S is not unique. The more physical object is the action difference between two path γ, γ', with the same initial starting point and end point. A small loop is formed with area Γ and boundary $\partial\Gamma = \gamma - \gamma'$, and normal vector pointing outside the area $\boldsymbol{\sigma}$, consider the flux through the loop

$$\begin{aligned}
\Phi &= -\iint_\Gamma \mathbf{u} \cdot d\boldsymbol{\sigma} \\
&= \iint_\Gamma \nabla \times \boldsymbol{\theta} \cdot d\boldsymbol{\sigma} \\
&= \oint_{\partial\Gamma} \boldsymbol{\theta} \cdot d\mathbf{r} \\
&= \oint_{\gamma-\gamma'} \boldsymbol{\theta} \cdot d\mathbf{r} \\
&= \int_\gamma \boldsymbol{\theta} \cdot d\mathbf{r} - \int_{\gamma'} \boldsymbol{\theta} \cdot d\mathbf{r}' \\
&= S_\gamma - S_{\gamma'} \\
&= \Delta S
\end{aligned} \tag{8}$$

If γ is along the phase flow $\mathbf{u} \cdot d\boldsymbol{\sigma} = 0$, then a small deviation path γ', with result in no flux $\Phi = \delta S = 0$, which is the principle of least action. So postulate Eq.(5) is consistent with Hamilton's equations and Principles of least action. We now make a connection to $S^{(3)}$, from definition Eq.(2) we have

$$dS^{(3)} = \frac{\partial S^{(3)}}{\partial \bar{p}} d\bar{p} + \frac{\partial S^{(3)}}{\partial \bar{q}} d\bar{q} = \Delta q \, d\bar{p} - \Delta p \, d\bar{q} \tag{9}$$

Now consider

$$S = \int_0^t (p\dot{q} - H) d\tau \tag{10}$$

$$\begin{aligned}
dS &= \int_0^t \left(\dot{q} \, dp + p \, d\dot{q} - \frac{\partial H}{\partial q} dq - \frac{\partial H}{\partial p} dp \right) d\tau \\
&= \int_0^t (p \, d\dot{q} + \dot{p} \, dq) \, d\tau \\
&= \int_0^t \frac{d}{d\tau} (p \, dq) \, d\tau \\
&= [p \, dq]_0^t \\
&= p' \, dq' - p \, dq
\end{aligned} \tag{11}$$

then after a bit of algebra we have

$$p' dq' - p dq = \bar{p} \, d\Delta q + \Delta p \, d\bar{q} \tag{12}$$

then it follows that

$$dS^{(3)} = d(\bar{p}\Delta q - S) \tag{13}$$

thus up to a physically irrelevant constant

$$S^{(3)} = \bar{p}\Delta q - S \tag{14}$$

To understand the geometry of $S^{(3)}$, consider a physical trajectory γ, which is $(q,p,0)\to(q',p',t)$, then use a straight chord c to make a second connection $(q,p)\to(q',p')$ to form a loop $\gamma - c$

$$c(s) = \begin{pmatrix} q_c(s) \\ p_c(s) \\ st \end{pmatrix} = \begin{pmatrix} q + s\Delta q \\ p + s\Delta p \\ st \end{pmatrix}, \qquad 0 \le s \le 1 \tag{15}$$

the midpoint is then $c(1/2) = (\bar{q}, \bar{p}, t/2)$. Consider the path integral along the chord

$$\int_c p\, dq = \int_0^1 p_c(s)\frac{dq_c}{ds}ds = \Delta q \int_0^1 (p + s\Delta p)ds = \bar{p}\Delta q \tag{16}$$

The loop in general is non planar, we project it onto (p,q) plane, its projected area is

$$A = \int_q^{q'} [p_\gamma(x) - p_c(x)]\, dx = \int_\gamma p\, dq - \int_c p\, dq = \oint_{\gamma - c} p\, dq \tag{17}$$

using Eq.(11) gives

$$\int_\gamma p\, dq = A + \bar{p}\Delta q \tag{18}$$

but since $\boldsymbol{\theta}\cdot d\mathbf{r} = p\, dq - H\, dt$, and $H = E$ is constant for γ, because it is a physical trajectory then

$$S_\gamma = A + \bar{p}\Delta q - Et \tag{19}$$

use Eq.(14)

$$\boxed{S^{(3)} = Et - A} \tag{20}$$

Since the signed area is anti-symmetric under time reversal, $S^{(3)}$ is an odd function in time. For multiple particles in three dimension, the area is

$$A = \oint_{\gamma - c} \mathbf{p}\cdot d\mathbf{q} = \sum_k \oint_{\gamma - c} p_k\, dq_k \tag{21}$$

Appendix A

It is easy to show Eq.(2) has the following solutions

$$q' = q\cos\omega t + \frac{p}{m\omega}\sin\omega t, \qquad p' = p\cos\omega t - m\omega q\sin\omega t \tag{A1}$$

Consider

$$\begin{aligned} \frac{2}{m\omega}\tan\left(\frac{\omega t}{2}\right)\bar{p} &= \frac{\sin\omega t}{m\omega(1+\cos\omega t)}(p + p') \\ &= \frac{p}{m\omega}\sin\omega t - \frac{\sin^2\omega t}{1+\cos\omega t}q \\ &= \frac{p}{m\omega}\sin\omega t + (\cos\omega t - 1)q \\ &= q' - q \\ &= \Delta q \end{aligned} \tag{A2}$$

where we used $\tan(\theta/2) = \sin\theta/(1+\cos\theta)$. Similarly by repeating the same steps

$$-2m\omega \tan\left(\frac{\omega t}{2}\right)\bar{q} = \Delta p \tag{A3}$$

thus Eq.(3) becomes

$$\frac{\partial S^{(3)}}{\partial \bar{p}} = \frac{2}{m\omega}\tan\left(\frac{\omega t}{2}\right)\bar{p}, \qquad \frac{\partial S^{(3)}}{\partial \bar{q}} = 2m\omega \tan\left(\frac{\omega t}{2}\right)\bar{q} \tag{A4}$$

integrating with respect to $\bar{p}$ and $\bar{q}$ and after a slight rearrangement gives

$$S^{(3)} = \frac{2}{\omega}\tan\left(\frac{\omega t}{2}\right)H(\bar{p},\bar{q}). \tag{A5}$$

Appendix B

i

The Hamilton's equation can be written as

$$\begin{pmatrix}\dot{q}\\ \dot{p}\end{pmatrix} = \begin{pmatrix}0 & 1\\ -1 & 0\end{pmatrix}\begin{pmatrix}\frac{\partial H(q,p)}{\partial q}\\ \frac{\partial H(q,p)}{\partial p}\end{pmatrix} \tag{B1}$$

define the 90 degrees clockwise rotation matrix in the middle as Ω, phase point $x = (q,p)$ and phase space gradient operator $\partial_x = (\partial_q, \partial_p)$, we see the phase space velocity is orthogonal to the direction of steepest ascent in energy, so clearly energy is conserved. We rewrite Hamilton's equation as

$$\dot{x} = \Omega\frac{\partial H}{\partial x} \tag{B2}$$

Now define the Jacobian of the transformation $(q,p) \to (q',p')$

$$J = \frac{\partial x'}{\partial x} = \begin{pmatrix}\frac{\partial q'}{\partial q} & \frac{\partial q'}{\partial p}\\ \frac{\partial p'}{\partial q} & \frac{\partial p'}{\partial p}\end{pmatrix} \tag{B3}$$

where prime mark $'$ denotes the phase point after it has been evolved for some time t. Now define Hessian

$$K = \frac{\partial^2 H(x')}{\partial x'^2} = \begin{pmatrix}\frac{\partial^2 H(x')}{\partial q'^2} & \frac{\partial H(x')}{\partial q'\partial p'}\\ \frac{\partial H(x')}{\partial p'\partial q'} & \frac{\partial H(x')}{\partial p'^2}\end{pmatrix} \tag{B4}$$

where $K = K^T$, then consider

$$\dot{J} = \frac{\partial^2 x'}{\partial t \partial x} = \frac{\partial^2 x'}{\partial x \partial t} = \Omega\frac{\partial}{\partial x}\frac{\partial H(x')}{\partial x'} = \Omega\frac{\partial^2 H(x')}{\partial x'^2}\frac{\partial x'}{\partial x} \tag{B5}$$

we have severely abused this ∂_x notation to reduce clutter on the page, but one can verify this by diffrentiating each component of the Jacobian individually. Now use definition of K and J gives

$$\dot{J} = \Omega K J \tag{B6}$$

Now consider

$$\frac{d}{dt}(J^T\Omega J) = \dot{J}^T\Omega J + J^T\Omega\dot{J} = J^T K^T\Omega^T\Omega J + J^T\Omega\Omega K J = J^T(K-K)J = 0 \tag{B7}$$

where we used $\Omega^2 = -I$, $\Omega^T = -\Omega$ and $K^T = K$ for the third equality, therefore the object $J^T\Omega J$ has no time dependence, then evaluate the expression at initial time where $(q', p') = (q, p)$ gives

$$J^T\Omega J = \Omega. \tag{B8}$$

Now we introduce jargon: A matrix J is called **symplectic** if it satisfies this relation.

ii

The Poisson bracket between f and g is

$$\{f, g\} = \frac{\partial f}{\partial q}\frac{\partial g}{\partial p} - \frac{\partial g}{\partial q}\frac{\partial f}{\partial p} \tag{B9}$$

for position and momentum the classical analogue of the canonical commutation relation is

$$\{q, p\} = 1 \tag{B10}$$

notice that

$$\{q', p'\} = \det J \tag{B11}$$

matrix multiplication gives

$$J^T\Omega J = \Omega \det J \tag{B12}$$

but J is symplectic, so the Poisson bracket is preserved.

$$\{q', p'\} = 1 \tag{B13}$$

this is analogous to $[q', p'] = [q, p] = i\hbar$ where $q' = q'(t) = e^{-itH/\hbar}qe^{itH/\hbar}$, $p' = p'(t) = e^{-itH/\hbar}pe^{itH/\hbar}$ in quantum mechanics. In fact we can write

$$\Omega = \begin{pmatrix} \{q, q\} & \{q, p\} \\ \{p, q\} & \{p, p\} \end{pmatrix} \tag{B14}$$

which is perhaps easier to interpret physically.

iii

We show the standard velocity verlet integrator is symplectic here. The velocity verlet has following three sub steps: half kick, drift, half kick

$$\bar{p} = p + \frac{\Delta t}{2}f(q), \qquad q' = q + \frac{\Delta t}{m}p, \qquad p' = \bar{p} + \frac{\Delta t}{2}f(q') \tag{B15}$$

one can show that the Jacobian of the half kick $K(q)$ and drift D are

$$K(q) = \begin{pmatrix} 1 & 0 \\ \frac{\Delta t}{2}\frac{\partial f}{\partial q} & 1 \end{pmatrix}, \qquad D = \begin{pmatrix} 1 & \frac{\Delta t}{m} \\ 0 & 1 \end{pmatrix} \tag{B16}$$

then by matrix multiplication, we have

$$K^T\Omega K = \Omega, \qquad D^T\Omega D = \Omega \tag{B17}$$

so even each sub step is symplectic, for the full step

$$K^T D^T K^T \Omega K D K = K^T D^T \Omega D K = K^T \Omega K = \Omega \tag{B18}$$

and it also follows that any amount of steps are also sympletic.